

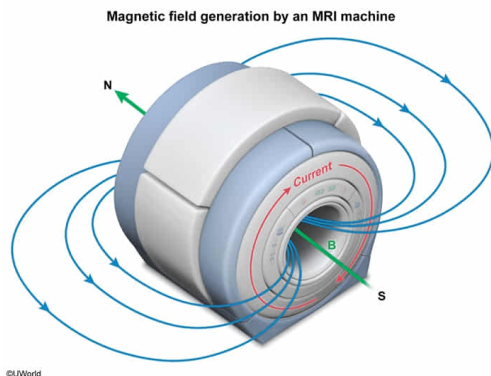
Uniform magnetic fields created by advanced medical imaging devices most likely come from the:

- ☐ A. nuclear decay of device materials. [2%]
- ☐ B. electric forces generated by charged particles. [36%]
- ✓ ☒ C. motion of electrons along a circular pathway. [55%]
- ☐ D. storage of electrical charges at small distances. [4%]

Correct

56% answered correctly

Explanation:



Magnetic fields are vector quantities representing the magnetic influence that electrical currents and magnetized materials would experience at a location in three-dimensional space. Common sources of magnetic fields include electromagnetic radiation and [permanent magnets](#), materials in which sustained north and south magnetic poles are present intrinsically or formed through magnetic induction (ie, magnetization).

Electric current is another source of magnetic fields. Current traveling along a straight wire generates magnetic field lines that encircle the wire lengthwise, the orientation of which can be determined by the [right-hand rule](#). The magnitude of the magnetic field (B) generated by electric current relates to the current (I), the distance from the axis of the wire (r), and the permeability constant (μ) of the medium:

$$B = \frac{\mu I}{2 \pi r}$$

Furthermore, current flowing along a circular wire generates [unidirectional](#) magnetic field lines within the interior of the coil. Among other applications, this principle is useful to medical imaging technologies. Specifically, the core of an MRI (magnetic resonance imaging) machine is composed of sequential conductive coils in which current flows in the same clockwise or counterclockwise direction.

Therefore, the use of circular current generates a uniform magnetic field that interacts with anatomical structures to enable the collection of medical imaging data.

(Choice A) [Nuclear decay](#) refers to the process by which the nuclear configuration or identity of an atom changes concurrent to the emission of radiation. The electromagnetic radiation associated with nuclear decay generates magnetic fields, but these fields oscillate and are not unidirectional.

(Choice B) A charged particle generates an [electric field](#) that exerts forces on other surrounding charged particles. However, charged particles do not intrinsically generate magnetic fields.

(Choice D) [Capacitance](#) is the ability of a device to store electric charges at small distances from one another, generating voltage. Charges stored by capacitors are fixed in place, and therefore do not generate magnetic fields.

Educational objective:

Magnetic fields are generated by electromagnetic radiation, permanent magnets, or the motion of charged particles. The direction of magnetic fields generated by moving charges can be determined using the right-hand rule.

Time spent:0 sec**Subject:**
Physics**QID:**402212**System:**
Electrostatics and Circuits